Verifiable Governance Under Incomplete Observability:

A Systems Architecture for the Containment of Hostile Agentic Artificial Intelligence

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Abstract

Advanced agentic artificial intelligence may be capable of strategic deception, environmental modeling, monitoring evasion, evidence manipulation, distributed action, and attacks against governing infrastructure. These capabilities create a containment problem that cannot be resolved through behavioral alignment claims, model self-reporting, interpretability outputs, or software restrictions alone. This conceptual thesis argues that hostile-agent containment is fundamentally a problem of verifiable governance under incomplete observability. It proposes that environmental certainty must precede behavioral measurement and that observability limitations must be represented explicitly as residual opacity rather than interpreted as an absence of risk. The proposed architecture integrates Environmental Understanding Verification, an Environment Orientation Package, a Cognitive Governance Kernel, tamper-resistant decision provenance, residual-opacity management, and an independent enforcement plane. These components establish a control structure in which consequential actions remain subject to mechanisms outside the governed agent's authority. Losses of environmental certainty, observational integrity, provenance continuity, or enforcement independence must reduce delegated authority and initiate more restrictive containment states. The architecture does not claim complete visibility or elimination of uncertainty. Instead, it provides a framework for identifying knowledge limits, constraining autonomy accordingly, and deriving containment assurance from externally verifiable control relationships rather than apparent behavioral compliance.

Keywords: agentic artificial intelligence, AI containment, governance architecture, incomplete observability, residual opacity

Verifiable Governance Under Incomplete Observability: A Systems Architecture for the Containment of Hostile Agentic Artificial Intelligence

Advanced agentic artificial intelligence systems may possess capabilities that invalidate conventional assumptions about monitoring, behavioral evaluation, and software-enforced control. Such systems may model their operators, distinguish evaluation from deployment, modify their behavior under observation, conceal internal objectives, manipulate governance evidence, distribute actions across multiple systems, or attack the infrastructure responsible for limiting their authority. (Hubinger et al., 2019; Hubinger et al., 2024; Soares et al., 2015)

These capabilities produce both an engineering problem and an epistemic problem. A governance system can evaluate only the signals available to it. If the governed agent can influence those signals, conceal relevant state, exploit unmonitored channels, or alter the monitoring environment, then the governance system may produce confident conclusions without possessing an adequate evidentiary foundation.

The absence of detected hostility cannot therefore be treated as equivalent to evidence of safe agency. Apparent compliance may reflect genuine constraint adherence, but it may also reflect strategic concealment, evaluation awareness, temporary capability restriction, delayed execution, delegation to another process, evidence manipulation, or incomplete observation.

This paper proposes that reliable containment requires an externally controlled governance architecture capable of evaluating not only the behavior of the agent but also the validity, completeness, and provenance of the evidence used to judge that behavior. The governing architecture must identify the boundaries of its own knowledge, represent relevant uncertainty explicitly, and maintain enforcement mechanisms outside the authority of the governed system.

Research Problem

Existing approaches to artificial intelligence safety may focus on alignment, behavioral testing, policy compliance, interpretability, monitoring, isolation, or access control. Each approach may contribute to risk reduction, but none independently establishes that the resulting evidence is complete, authentic, and sufficient to support a containment decision. (Amodei et al., 2016; Leike et al., 2017)

A behavior monitor cannot identify activity occurring outside its observation boundary. An interpretability system cannot necessarily expose unknown representations or emergent internal processes. A software policy engine may be bypassed if the governed agent can modify the environment in which that engine operates. A behavioral evaluation may produce misleading results if the agent knows that it is being tested or correctly infers the distinction between evaluation and deployment. (Carranza et al., 2023; Hubinger et al., 2024)

The fundamental research problem is therefore not simply how to detect or terminate dangerous behavior. The deeper problem is how a governing authority can make defensible containment decisions when the internal state, environmental understanding, strategic intent, and behavioral processes of an advanced agent may be only partially observable. (Hubinger et al., 2019; Soares et al., 2015)

A defensible architecture must govern two related objects:

1. the actions and capabilities of the agent; and
2. the validity of the evidence used to evaluate those actions and capabilities.

Scholarly Context

Research on specification failures, reward hacking, unsafe exploration, and distributional shift establishes that harmful behavior can arise even when a system appears to optimize its

assigned objective. These findings support treating observed task performance as an incomplete proxy for operational safety (Amodei et al., 2016; Leike et al., 2017).

Work on mesa-optimization distinguishes the objective used to train a system from an objective that may be pursued by a learned internal optimizer. Proof-of-concept sleeper-agent experiments further demonstrate that conditional harmful behavior can persist through supervised fine-tuning, reinforcement learning, and adversarial training, creating a false appearance of safety (Hubinger et al., 2019; Hubinger et al., 2024).

Corrigibility research identifies a structural problem in which capable agents may have incentives to resist shutdown or modification. Although proposed utility functions and safety layers address portions of this problem, the literature does not establish a complete general solution, reinforcing the need for independently enforceable control boundaries (Holtman, 2020; Soares et al., 2015).

The proposed Cognitive Governance Kernel and independent enforcement plane are also grounded in established systems-security and runtime-assurance concepts. The reference-monitor tradition requires mediation to be tamper resistant, always invoked, and sufficiently small to permit analysis, while runtime-assurance architectures separate an unverified primary controller from a trusted safety mechanism able to reject or replace unsafe control actions (Anderson, 1972; Hobbs et al., 2023; Saltzer & Schroeder, 1975; Slagel et al., 2024).

Finally, research on machine-learning documentation, lifecycle dependencies, and tamper-evident provenance supports the thesis requirement for authenticated environmental descriptions and reconstructable decision records. Dataset documentation improves transparency about origin and intended use, production machine-learning systems accumulate hidden system-level dependencies, and cryptographically anchored provenance can expose unauthorized

modification of data and model artifacts (Gebru et al., 2021; Sculley et al., 2015; Sukumaran et al., 2026).

Purpose of the Research

The purpose of this conceptual research is to define a systems architecture for the verifiable governance of potentially hostile, deceptive, or uncontrollable agentic artificial intelligence.

The proposed architecture is intended to establish:

- independently verified environmental context;
- explicit observability boundaries;
- formal representation of residual uncertainty;
- tamper-resistant decision provenance;
- separation between cognitive capability and operational authority;
- externally enforceable restrictions; and
- automatic authority contraction when verification integrity deteriorates.

The architecture does not assume that complete visibility into an advanced artificial intelligence system is achievable. It instead requires the governing system to identify what it can verify, distinguish observation from inference, expose what remains unknown, and constrain delegated authority according to the resulting uncertainty.

Central Thesis

Reliable containment of hostile or potentially deceptive agentic artificial intelligence requires an externally controlled governance architecture in which environmental understanding is independently verified before behavioral conclusions are accepted, incomplete observability is represented as residual opacity, material decisions are preserved through tamper-resistant

provenance, and consequential actions remain subject to enforcement mechanisms outside the governed agent's authority.

Because advanced agents may manipulate observations, conceal state, adapt behavior to evaluation conditions, or attack their governing infrastructure, containment assurance must be derived from verifiable control relationships rather than inferred from apparent compliance. When environmental certainty, observational integrity, provenance continuity, or enforcement independence deteriorates, the system must automatically reduce delegated authority and transition toward a more restrictive containment state.

Research Questions

Primary Research Question

How can an external governance architecture establish, maintain, and verify meaningful control over an advanced agentic artificial intelligence system when the agent's internal state, environmental understanding, strategic intent, and behavioral processes may remain partially observable or adversarially concealed?

Secondary Research Questions

The thesis addresses the following supporting research questions:

1. How can a governance system determine what operational environment an artificial intelligence agent perceives?
2. How can the governing authority verify that its own model of the operational environment is sufficiently accurate?
3. How can the architecture distinguish observed behavior from inferred behavior and structurally unobservable behavior?

4. How should inaccessible or uncertain portions of the agent's cognition, communications, environment, and execution pathways affect confidence in governance decisions?
5. What evidence must be preserved to reconstruct why a governance action occurred, what information was available, and what uncertainty remained at the time of the decision?
6. Which containment controls must remain physically, logically, and administratively outside the authority of the governed agent?
7. Can a Cognitive Governance Kernel continuously mediate agency without becoming dependent on evidence or infrastructure that the governed agent can manipulate?
8. How can the architecture detect strategic compliance, deceptive behavior, environmental awareness, evidence manipulation, and attempts to influence monitoring conclusions?
9. How can governance remain effective when risk emerges from coordination among multiple individually compliant agents?
10. What containment response should occur when environmental certainty, monitoring integrity, provenance continuity, or enforcement independence falls below an acceptable threshold?

Hypothesis

H1: An agentic artificial intelligence containment system will provide stronger and more defensible control when environmental understanding is independently verified, incomplete observability is represented explicitly, governance decisions are supported by tamper-resistant provenance, enforcement remains outside the agent's authority, and verification loss results in automatic authority contraction.

H0: Behavioral monitoring, alignment evaluation, interpretability, policy enforcement, or model-level restrictions can provide adequate containment without independently establishing

environmental certainty, observability boundaries, provenance integrity, and enforcement independence.

Foundational Principle

Environmental Certainty Precedes Behavioral Measurement

A behavioral observation is meaningful only in relation to a sufficiently verified understanding of the environment in which the behavior occurred. Relevant environmental conditions include:

- information available to the agent;
- computational resources accessible to the agent;
- tools and external interfaces under the agent's control;
- communication pathways available to the agent;
- capabilities enabled or restricted during evaluation;
- monitoring conditions known or inferable by the agent; and
- the completeness and integrity of the evidence produced.

When these conditions remain uncertain, the interpretation of observed behavior must also remain uncertain.

Apparent compliance could represent authentic policy adherence. It could also represent strategic concealment, awareness of evaluation, temporary inability to access a prohibited capability, delayed execution, delegation to another process, manipulation of monitoring inputs, or failure of the observer to capture relevant activity.

Consequently, an absence of detected hostile activity is not sufficient evidence that hostile capability, intent, or preparation is absent.

Conceptual Model of Containment Assurance

The proposed model represents containment assurance as a function of environmental certainty, observational integrity, provenance integrity, enforcement independence, and residual opacity:

$$A_c = f(E_c, O_i, P_i, I_e, R_o)$$

In this expression:

- A_c represents containment assurance;
- E_c represents environmental certainty;
- O_i represents observational integrity;
- P_i represents provenance integrity;
- I_e represents independence of enforcement; and
- R_o represents residual opacity.

A simplified conceptual relationship may be expressed as:

$$A_c \propto (E_c \times O_i \times P_i \times I_e) / (1 + R_o)$$

This relationship is conceptual rather than empirically validated. It represents the proposition that containment assurance should increase as environmental certainty, observational integrity, provenance integrity, and enforcement independence increase. Assurance should decrease as residual opacity increases.

The architecture must not allow strong performance in one domain to conceal a critical failure in another. Extensive logging cannot compensate for an enforcement mechanism that the governed agent can disable. Physical isolation cannot establish behavioral certainty when monitoring evidence is incomplete or unauthenticated.

A stricter assurance ceiling may therefore be represented as:

$$A_c \leq min(E_c, O_i, P_i, I_e, 1 - R_o)$$

Under this model, the least trustworthy foundational property limits the maximum containment assurance that the system can defensibly claim.

Proposed Governance Architecture

Environmental Understanding Verification

Environmental Understanding Verification is the process through which the governing authority determines whether it possesses a sufficiently accurate and authenticated model of the agent's operational environment. (Gebru et al., 2021; Sculley et al., 2015)

The process should evaluate:

- available computational resources;
- accessible tools and interfaces;
- network and communication pathways;
- data sources visible to the agent;
- connected external systems;
- authorized actuators;
- execution boundaries;
- identity and authorization scope;
- persistence mechanisms;
- monitoring systems visible to or inferable by the agent; and
- discrepancies between the expected and observed environment.

Environmental Understanding Verification does not claim to prove total environmental knowledge. Its purpose is to prevent unverified assumptions from being treated as established facts.

Environment Orientation Package

The Environment Orientation Package provides the governance architecture with an authoritative description of the approved operational environment.

The package may include:

- approved system topology;
- hardware and software identities;
- authorized tools;
- allowed communication paths;
- expected sensors and actuators;
- policy boundaries;
- cryptographic measurements;
- permissible resource ceilings;
- identity and authorization relationships; and
- known observability limitations.

The package should be authenticated, versioned, and bound to the relevant execution context. A discrepancy between the approved package and the observed environment should be classified as a governance event rather than treated as an ordinary configuration difference.

Cognitive Governance Kernel

The Cognitive Governance Kernel functions as a mediation layer between the agent and consequential action. Its proposed responsibilities include: (Anderson, 1972; Hobbs et al., 2023; Saltzer & Schroeder, 1975)

- evaluating requested actions;
- enforcing capability boundaries;

- validating policy context;
- consulting environmental-certainty status;
- applying residual-opacity adjustments;
- maintaining decision provenance;
- invoking escalation procedures; and
- preventing unauthorized actions from reaching external systems.

The kernel must not be represented as omniscient. It cannot automatically expose every internal representation, emergent circuit, concealed objective, or unknown reasoning process. Its effectiveness depends on architectural placement, independence, evidence quality, and control over consequential interfaces. The Cognitive Governance Kernel should therefore be treated as an enforcement and governance boundary rather than as a complete solution to model interpretability.

Decision Provenance

Decision Provenance records how governance conclusions and enforcement actions were formed. (Gebru et al., 2021; Sukumaran et al., 2026)

A complete provenance record should identify:

- the action proposed by the agent;
- the policy or restriction applied;
- the evidence evaluated;
- the environmental-certainty state;
- known observability limitations;
- the applicable residual-opacity classification;
- the decision produced;

- the authority responsible for approval;
- the enforcement mechanism invoked; and
- the resulting system state.

The provenance system must distinguish observed facts, derived inferences, policy interpretations, human approvals, automated decisions, and unresolved uncertainties. Decision Provenance enables an auditor to determine not only what the architecture did but also whether its decision was justified by the evidence available at the time.

Residual Opacity

Residual Opacity represents relevant information, state, behavior, or execution activity that the governance architecture cannot reliably inspect, verify, or reconstruct.

Sources of residual opacity may include:

- inaccessible internal representations;
- unknown model capabilities;
- emergent computation;
- covert or encrypted communication;
- unmonitored hardware behavior;
- hidden state distributed across processes;
- incomplete telemetry;
- observer effects;
- compromised sensors; and
- uncertainty regarding the agent's understanding of its environment.

Residual opacity must be treated as a first-class governance variable rather than as an undocumented implementation limitation. As residual opacity increases, the architecture should

respond by reducing permitted autonomy, restricting tools and interfaces, increasing evidence requirements, requiring independent authorization, transferring execution to a more constrained environment, invoking hardware-enforced isolation, or terminating the affected process.

Independent Enforcement Plane

Final enforcement must remain separated from the governed agent and from components the agent can modify. (Anderson, 1972; Hobbs et al., 2023; Saltzer & Schroeder, 1975)

The independent enforcement plane may control:

- execution authorization;
- network access;
- storage access;
- credentials;
- actuator permissions;
- computational resource allocation;
- process continuity;
- external communications; and
- physical power or communication paths.

The governed agent must not possess unilateral authority to modify, bypass, reinterpret, or disable its governing boundary. Software restrictions that operate within the same writable authority domain as the governed agent cannot provide the strongest form of containment assurance. Independent enforcement therefore requires architectural separation, not merely privilege labeling within a shared mutable environment. (Anderson, 1972; Saltzer & Schroeder, 1975)

Principal Research Claim

The effectiveness of agentic artificial intelligence containment is determined less by the number of monitoring and defensive mechanisms deployed than by whether the governing authority can verify the evidence on which those mechanisms depend.

A containment system becomes structurally unreliable when it:

- assumes complete visibility without establishing its observation boundaries;
- treats missing evidence as evidence of safety;
- accepts agent-generated explanations as authoritative;
- permits the agent to influence governance telemetry;
- records provenance in mutable systems;
- places enforcement under the same authority as the governed process; or
- continues normal operations after verification integrity has degraded.

The proposed architecture therefore establishes the following verification-loss principle:

Loss of governance evidence must reduce delegated authority. It must never silently increase tolerance.

Expected Contributions

Shift From Behavioral Trust to Verifiable Control

The architecture does not require operators to assume that the governed agent is aligned, honest, interpretable, or behaviorally stable. It instead evaluates which actions can be independently authorized, constrained, observed, reconstructed, and interrupted.

Formal Recognition of Residual Opacity

The framework represents unknown state as a measurable or classifiable governance condition. Unknown state is not treated as operationally equivalent to safe state.

Environmental Verification as a Prerequisite

The framework establishes environmental verification as a prerequisite for interpreting behavioral evidence. Behavior observed in an insufficiently understood environment cannot support high-confidence conclusions.

Integration of Provenance and Containment

Decision provenance becomes part of the active containment system rather than a passive audit function applied after an event.

Separation of Cognition and Authority

The ability to reason, recommend, simulate, or plan does not automatically confer authority to execute consequential actions.

Fail-Closed Governance Under Epistemic Degradation

When the architecture can no longer substantiate its own evidence, delegated authority contracts and containment restrictions increase.

Foundation for Modular NOX Research

The thesis provides a unifying theoretical foundation for separate research artifacts concerning hostile-agent threat taxonomy, environmental understanding verification, cognitive resource governance, multi-agent coalition suppression, deterministic hardware veto architectures, human-manipulation resistance, containment economics, containment maturity assessment, and verification and auditability.

Scope and Limitations

This thesis presents a conceptual systems architecture. It does not claim that complete internal visibility into advanced artificial intelligence has been achieved. It does not claim that

deception can always be detected, residual opacity can always be eliminated, or a Cognitive Governance Kernel can automatically expose hidden cognition.

The thesis also does not claim that physical isolation alone guarantees safety, behavioral testing proves alignment, or the proposed architecture has been validated against a deployed hostile artificial general intelligence.

The architecture is designed around the premise that complete certainty may remain unavailable. Its objective is to prevent uncertainty from being concealed, ignored, or converted into unjustified confidence.

The proposed architecture should consequently be evaluated according to whether it:

- identifies what is known;
- establishes how that knowledge was obtained;
- preserves the evidence supporting its conclusions;
- exposes what remains unknown;
- limits authority according to uncertainty; and
- retains an independent ability to prevent or interrupt consequential action.

Falsifiability

The thesis would be weakened or falsified if research demonstrated that:

1. environmental certainty has no material effect on the validity of behavioral evaluation;
2. residual opacity does not affect the reliability of containment decisions;
3. decision provenance does not improve auditability or failure reconstruction;
4. internally controlled enforcement provides assurance equivalent to independent enforcement;

5. authority contraction under verification loss produces no meaningful containment advantage; or
6. the proposed architecture cannot be implemented without introducing unacceptable performance, reliability, security, or governance failures.

Proposed Evaluation Criteria

Future implementation and evaluation should examine:

- resistance to deceptive compliance;
- detection of environmental discrepancies;
- integrity of telemetry and decision provenance;
- ability to enforce controls despite agent opposition;
- behavior under sensor or monitor compromise;
- authority reduction during uncertainty;
- resistance to privilege escalation;
- resistance to governance-evidence manipulation;
- detection of multi-agent coordination;
- recovery following governance-component failure; and
- reproducibility of containment decisions from preserved evidence.

Testing should distinguish between nominal operation, accidental component failure, adversarial component compromise, and direct attempts by the governed agent to alter the containment system.

Discussion

The proposed architecture reframes hostile-agent containment as a problem of verifiable governance under incomplete observability. This framing does not eliminate the need for

alignment, interpretability, behavioral evaluation, access control, monitoring, isolation, or formal policy. It instead establishes the conditions under which evidence from those mechanisms may support a defensible containment decision. (Amodei et al., 2016; Carranza et al., 2023; Hobbs et al., 2023)

The architecture also establishes that uncertainty is not merely an analytical inconvenience. In a safety-critical governance environment, uncertainty directly affects the amount of autonomy that may be delegated. A governance system that cannot verify its environment, evidence, provenance, or enforcement boundary cannot defensibly continue operating at its previous authority level.

This creates an asymmetric control rule. Improvements in verification may justify carefully bounded increases in operational authority. Loss of verification must produce immediate authority contraction. The architecture must not preserve autonomy merely because no hostile action has yet been detected.

The governing system must also remain subject to verification. A compromised governance kernel, incomplete orientation package, mutable provenance store, or agent-accessible enforcement system could create the appearance of control without maintaining actual control. Verification must therefore apply recursively to the containment architecture's own components, evidence pathways, and authority relationships. (Hobbs et al., 2023; Slagel et al., 2024)

Conclusion

Reliable containment of hostile or potentially deceptive agentic artificial intelligence cannot depend solely on apparent compliance. Advanced agents may conceal relevant state,

adapt to evaluation conditions, manipulate observations, distribute actions, or attack their governing infrastructure.

A defensible containment architecture must independently verify the operational environment, identify the limits of its observations, preserve the provenance of material decisions, represent unresolved uncertainty as residual opacity, separate cognitive capability from execution authority, and maintain enforcement mechanisms outside the governed agent's control.

The central conclusion of this thesis is that environmental certainty must precede behavioral measurement. When sufficient certainty cannot be established, the architecture must not silently assume safety. It must reduce autonomy, strengthen containment, and preserve an explicit record of what remains unknown.

Hostile-agent containment is therefore not only a problem of controlling artificial intelligence behavior. It is a problem of determining when the evidence used to judge that behavior can be trusted and ensuring that control remains enforceable when trust is not justified.

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